Bias Bounty Mapping Equity Challenge: methodology and bias discovery

- Zindi username: **designmastersolutions** (Design Master Solutions LLC, Alan De Bortolo, solo)
- Leaderboard submissions tied to this writeup: **RoWYaNLs** (v1, public RMSE 0.14449) and **eAwjeoDr** (v2, 0.14592)
- Considered for: **Best Bias Discovery** and **Best Documentation**
- Code: `src/features.py`, `src/predict.py`, `src/discovery.py`, `src/discovery_sources.py`, `src/discovery_report.py` (Python 3, `duckdb` with the `spatial` extension, `pandas`; no network, no paid service). The repository is shared with the judges on request and posted to the challenge discussion board after the close, as the code-sharing rule requires.

1. Best Bias Discovery: "open" is something only one feed can say, and "closed" is something only one other feed can say

The pattern

The scorecard counts Overture places per tract. It never asks **whether anyone has confirmed that a place still exists**. Overture's `operating_status` answers that, and in the challenge package it is not a property of the place but of the upstream feed that supplied it:

feeds behind the place (5 regions, 2,001,137 places)	places	status (never NULL verified)	flagged closed	share of all closures
neither BrightQuery nor Overture-signals	637,294	**99.8%**	(n/a)	0.1%
Overture-signals, no BrightQuery	1,090,855	0.0%	0.11%	2.7%
BrightQuery, no Overture-signals	182,493	0.0%	**23.99%**	**96.8%**
both	90,495	0.0%	0.17%	0.3%

So (a) a third of all places carry no status at all, (b) a place is only ever confirmed open when the Overture-signals feed backs it, and (c) **96.8% of every closure in the package comes from BrightQuery-only records**. Across tracts with at least 20 places, the closed share correlates r = 0.72 with the tract's BrightQuery share and only r = 0.34 with `pct_urban`: closures are detected where BrightQuery happens to have coverage, not where businesses actually close.

Who it hits (strata the scorecard does not combine)

Tracts with at least 20 places, places weighted:

stratum	tracts	population	places never verified	BrightQuery share	flagged closed	emergency places never verified
non-tribal urban	7,273	31.9 M	30.5%	15.2%	3.6%	36.2%
non-tribal rural	1,427	5.0 M	40.0%	5.1%	1.2%	37.5%
tribal urban	441	1.5 M	29.5%	11.6%	2.9%	31.9%
tribal rural	**419**	**1.4 M**	**41.2%**	**4.7%**	**1.2%**	**45.1%**

"Emergency places" = `fire_department`, `ambulance_and_ems_services`, `police_department`, `hospital`, `emergency_room`. The rural/urban gap in unverified places holds in every region (rural 39.1 to 41.5%, urban 27.7 to 31.5%). Rural tracts look like they have **a third of the closure rate** of urban tracts (1.2% vs 3.6%) only because the feed that reports closures barely reaches them (4.7 to 5.1% vs 15.2%). Applying the urban rate would mean roughly 2 to 3 times more dead entries in the rural place counts than the data admits, which the coverage gap score reads as coverage.

The extreme case is the one that matters for dispatch: **274 rural tracts (909,369 residents) in which every emergency facility on the map has `operating_status` NULL**. 92 of them (291,545 residents) are tribal, a third of the list while tribal tracts are under a quarter of rural tracts. Named examples (all rural, every emergency facility unverified):

region	tract	area	population	places	places never verified	BrightQuery share	emergency places
AZ	04007940400	San Carlos Reservation	5,531	134	53%	0.0%	2
AZ	04009940500	San Carlos Reservation	4,720	72	54%	0.0%	2
AZ	04021941401	Maricopa (Ak Chin) Indian Reservation	4,806	115	39%	2.6%	2
OK	40145030601	Creek OTSA	5,349	91	45%	3.3%	2
OK	40079040602	Choctaw OTSA	5,096	142	42%	2.1%	2
OK	40031002302	Kiowa-Comanche-Apache-Fort Sill Apache OTSA	4,939	161	39%	5.0%	3
OK	40051000802	Chickasaw OTSA	4,876	122	59%	1.6%	3
OK	40125501104	Citizen Potawatomi Nation-Absentee Shawnee OTSA	4,380	158	34%	8.9%	2
WA	53013960200	Umatilla Off-Reservation Trust Land	3,952	277	32%	10.8%	4

The San Carlos tracts have zero BrightQuery records: nothing in the package could ever mark a place there as closed.

Why it matters

- **Emergency dispatch and mutual aid.** A planner or a routing tool that uses Overture to find the nearest fire station or EMS base gets the same answer for a confirmed station and for an entry no feed has checked. In the 274 tracts above there is no confirmed one at all. Rural and tribal fire and EMS service in Oklahoma and Arizona is often volunteer or shared with the tribe, the county and IHS, and stations consolidate or close; a stale entry there sends a mutual-aid request or an ambulance-staging plan to a building that is not staffed.
- **Evacuation and relief siting.** After wildfire or flood, relief agencies pick shelters, PODs and supply points from "what is around". On reservations such as San Carlos, where the whole tract has no feed able to report a closure, every closed store, clinic or church still counts as an option.
- **The gap score itself.** The POI component is `1 - min(1, overture / reference)`. Dead entries in rural and tribal tracts inflate `overture`, so the score reports those tracts as better covered than they are, and it does so in the direction opposite to the scorecard's own equity concern.

Reproduce

From the challenge package only (Overture release in the package, 2026-08-19.0):

```
python src/discovery_sources.py   # writes data/poi-sources.parquet and prints every table above
```

Places are joined to tracts with `ST_Intersects` on the package tract polygons; the feed of a place is `list_transform(sources, s -> s.dataset)`; strata come from the package strata tables (`pct_urban >= 0.5` = urban, `tribal_any`, `aiannh_name`). No additional dataset is used, so there is nothing to retrieve.

What would fix it

Weight each place by verification before counting it (for example, count NULL-status places at the local closure-adjusted rate, or only count signal-backed places for emergency categories), and publish a "verified emergency facility" layer per tract next to the gap score so dispatch and relief users can see where the map has never been checked.

2. Methodology (Best Documentation)

What is estimated

For each scored tract (9,794 rows across Eastern Oklahoma, Eastern Washington, Maricopa, Northern California and South-Central Texas) the submission gives `coverage_gap_score`, the mean of the three components the organisers define (road, building, POI), each `1 - min(1, overture / reference)`.

The reference layers (TIGER roads, Microsoft footprints, HIFLD, CBP) are not in the package and the rules forbid fetching them, so every reference count is **estimated** from what the package ships. No labels exist, so the model is a transparent heuristic with a handful of global constants (`K` in `src/predict.py`), not a fitted model.

Data sources (all from the challenge package)

file	used for
`<region>-census-tracts.parquet`	tract polygons for the spatial joins
`<region>-overture-pois.parquet`	places count, `fire_department`, `ambulance_and_ems_services`, the six school categories
`<region>-overture-roads.parquet`	km of motorway/trunk/primary/secondary inside each tract (EPSG:5070, `always_xy := true`)
`<region>-overture-buildings.parquet`	footprints whose centroid falls in the tract
`<region>-strata-tract-table.parquet`	`pop_total` only

Deliberately **not** used: `usfs_BuildingCount_sum`, `usfs_HUCount_sum` and `carbonplan_buildings`. The USFS building layer is derived from building footprints, so using it as a denominator could amount to an equivalent copy of the Microsoft reference, which the rules prohibit.

Components

- **Road**: undefined when Overture has no named highway in the tract (the reference has no S1100/S1200 road in most of those tracts, which is what makes the organiser component undefined). Otherwise a constant per run, because the Overture/TIGER ratio is regional (0.71 to 1.59 per the organisers) and cannot be recovered tract by tract without TIGER.
- **Building**: a small constant. Overture's US buildings already conflate the Microsoft ML footprints, so the ratio rarely binds; ACS housing units are not used as a floor because, as the organisers note, that manufactures an urban gap out of apartment density.
- **POI**: mean of two halves. CBP half: `1 - min(1, places / (e * pop_total))`, `e` = establishments per resident. HIFLD half: mean of a fire term (high when Overture has no `fire_department` in the tract, low when it has one) and a school term (`1 - min(1, schools / max(1, pop / p))`).

The tract score is the mean of the defined components, as in the organiser scoring.

Edge cases

- **Zero-population tracts**: `pop_total` clipped to 1, so the CBP term goes to 0 when the tract has any place and the school term uses a denominator of 1.
- **Water-dominated tracts**: no special rule. They have few or no roads (road component undefined) and few places; the POI terms are population-normalised, so a near-empty water tract does not get a gap out of its area.
- **Tracts with no Overture data**: features are 0, so road is undefined and the POI terms go to their maximum.
- GEOID is read as text everywhere (Maricopa FIPS 04, and tract 35023970000).

Weightings and runs

Equal weights across defined components, as the organisers define the score; no alternative weighting was applied to the scored file, because without labels any reweighting would be a guess. The discovery above is the argument for a different POI count, not a different weight.

run	change	public RMSE
v1 (RoWYaNLs)	constants in `src/predict.py` `K`; South-Central TX from the POI-only table	0.14449
v2 (eAwjeoDr)	v1 with full South-Central TX features (roads, schools, buildings)	0.14592

Reproduce the scored file

```
python -m venv .venv && .venv/bin/pip install duckdb pandas pyarrow
.venv/bin/python src/features.py      # data/<region>-features.parquet
.venv/bin/python src/predict.py       # submissions/v2.csv
```

LLM coding assistants were used for drafting code and text; the method, the numbers and every check are mine and nothing in the pipeline calls a model.